

COMPETITION BETWEEN FOUR-WAVE MIXING AND AMPLIFIED SPONTANEOUS EMISSION

Michelle S. MALCUIT, Daniel J. GAUTHIER and Robert W. BOYD

Institute of Optics, University of Rochester, Rochester, New York 14627, U.S.A.

There has recently been considerable interest in nonlinear optical interactions in which two different processes compete with one another. In this paper, we review some recent work in this area, and describe the results of our own investigation of competition between four-wave mixing and amplified spontaneous emission. We find that under two-photon excitation of the sodium 3d level, new optical frequency components can be generated either by amplified spontaneous emission at the 3d $\rightarrow$ 3p transition frequency or by a resonantly enhanced four-wave mixing process. We have observed competition between these two processes, resulting in the suppression of amplified spontaneous emission. The transfer of population to the 3d level is inhibited by the destructive interference between two different pathways connecting the ground and upper levels.

1. Introduction

In many optical systems, several different nonlinear optical processes can occur simultaneously; typically these processes do not evolve independently but tend to compete with one another in a complex and highly nonlinear manner. One of the earliest examples of competition effects in nonlinear optics is the competition that occurs between different stimulated scattering processes. Kaiser and Maier [1] noted that in a nonlinear material such as carbon disulfide, both stimulated Raman scattering and stimulated Brillouin scattering can occur. However it has been observed that for relatively long excitation pulses only stimulated Brillouin scattering occurred, whereas for a relatively short excitation pulse only stimulated Raman scattering occurred. The mechanism responsible for this competition was that the Brillouin gain in steady state is larger than the gain for stimulated Raman scattering, allowing the Brillouin signal to grow more rapidly and deplete the energy of the pump beam, which prevents the stimulated Raman scattering from occurring. In contrast, in the limit of short-pulse excitation, the transient gain for the Raman process is higher which does not allow the stimulated Brillouin scattering to build up.

In this paper we report on our work on competition between amplified spontaneous emission and the four-wave mixing (FWM) process [2,3]. In our experiment we tuned a dye laser to the 3s $\rightarrow$ 3d two-photon-allowed transition in atomic sodium vapor and under a broad range of experimental conditions

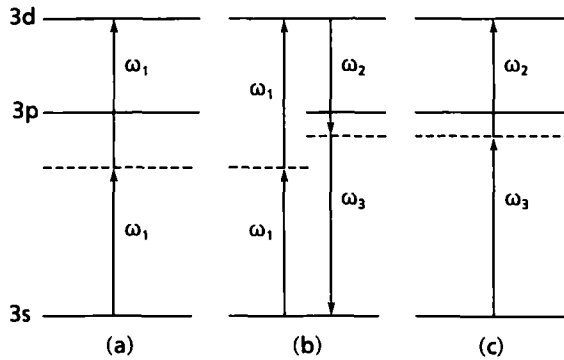


Fig. 1. The nature of the suppression of the upper level population by the FWM process is illustrated. (a) The incident laser field creates an excitation pathway connecting the ground and excited states. (b) Due to the FWM process, fields at frequencies ω_2 and ω_3 are generated. (c) These new fields create a second excitation pathway connecting these levels. Under quite general conditions, this pathway interferes destructively with that due to the incident field.

observed a strong FWM signal and no ASE. This result is surprising because the gain of the ASE, calculated neglecting the competition effects, is much larger than for the FWM process. We verified that the ASE was suppressed by the FWM process by exciting the medium with counterpropagating pump waves such that the FWM process could not be efficiently excited and observed strong ASE. The origin of the competition is illustrated in fig. 1. Initially two-photon absorption of the laser field is the only pathway of excitation connecting the ground and excited state since only the laser frequency ω_1 is applied to the cell. However FWM leads to the generation of new frequencies at ω_2 and ω_3 as shown in fig. 1b. These new fields create a second pathway of excitation connecting the ground and excited states involving two-photon absorption of a ω_2 and ω_3 photon, as shown in fig. 1c. We have shown that under a broad range of conditions the generated fields evolve in such a way that these two excitations pathways interfere destructively, thus preventing the excitation of the excited state. We have characterized the competition by solving perturbatively the density matrix equations of the atomic system. We find that when the wavevector mismatch is not too large the fields evolve spatially to reach spatially-invariant values and that the population in the 3d level is smaller than that due to the incident laser field alone.

The type of competition we have observed is an example of a coherent (that is, phase-matched) nonlinear optical process suppressing an incoherent nonlinear optical process. The first reported observation of such a competition effect was the observation of Miller et al. [4] and Glowina and Sander [5] of suppression of resonant multiphoton ionization by third-harmonic generation. Aron and John-

son [6] had obtained a rather surprising result in their study of the excitation spectrum for multiphoton ionization of xenon. They failed to detect the expected peak that corresponds to the $5p \rightarrow 6s$ three-photon resonance of xenon. In addition, another surprising experimental observation was obtained by Faisal et al. [7] in their study of three photon excitation of xenon. They observed that the fluorescence from the $6s$ level decreased for pressures greater than ~ 8 mTorr. To investigate these effects, Miller et al. [4] studied three-photon resonant, five-photon ionization of the $6s$ level of xenon while monitoring the emission from the atomic vapor. They found that for pressures above ~ 0.3 Torr of xenon the ionization signal disappeared and that simultaneously, intense ultraviolet light was emitted in the forward direction with a frequency equal to the third-harmonic of the applied field. This observation suggested that the intense generated field suppresses the ionization signal. To confirm that the third harmonic generation suppressed the multiphoton ionization, Glowia and Sander [5] measured the photoionization signal under conditions when third harmonic generation was forbidden, and compared the results to the case when third harmonic generation was strong. When circularly polarized, counterpropagating pump beams were used so that third harmonic generation was forbidden a strong multiphoton ionization signal was detected corresponding to the $6s$ resonance. However for linearly polarized copropagating pump beams no photoionization signal was detected whereas an intense third harmonic signal was detected. This experiment confirmed that the photoionization was suppressed by the third harmonic generation.

There have been several additional experimental and theoretical studies of this type of competition. Jackson et al. [8,9] and Wynne [10] demonstrated that the suppression of the photoionization was due to an interference effect between two excitation pathways of the excited state. They found that the generated field can evolve with amplitude and phase such that the pathway of excitation due to one-photon absorption of the generated field exactly cancels the three-photon absorption of the applied field. This interference results in diminished excitation of the excited state, and therefore a decrease in the resonant photoionization signal. Their theoretical treatment predicted that these types of competition effects could occur in any system in which the generated field makes a one-photon resonance in the medium. Agarwal and Tewari [11] have developed a quantum theory of competition between third harmonic generation and multiphoton ionization. Since these first observations of competition between third-harmonic generation and multiphoton ionization there have been many theoretical and experimental studies of this effect [12–18].

Another example of this type of interference has been studied recently by Krasnikov et al. [19]. They studied a four-wave mixing process in atomic sodium vapor in which two applied fields of frequencies ω_1 and ω_2 were tuned such that

the sum of their frequencies was equal to the $3s \rightarrow 4s$ transition frequency of atomic sodium. A third laser field of frequency ω_3 was applied and the signal at frequency $\omega_4 = \omega_1 + \omega_2 + \omega_3$ was measured as a function of atomic number density. They found that the conversion efficiency of the signal wave saturated at number densities greater than $\sim 3 \times 10^{15} \text{ cm}^{-3}$, and attributed the saturation to competition between two-photon absorption of the ω_1 and ω_2 waves and the two-photon absorption of the ω_3 and generated field at ω_4 . This competition was discussed previously by Manykin et al.[20] and is similar to the effect discussed by Kildal and Brueck [21] that they used to explain the anomalously low conversion efficiency they observed in two-photon resonant, third-harmonic generation.

From these examples it is clear that competition between coherent and incoherent processes is very general, and can occur in many experimental systems. Any process, such as ASE or photoionization, that requires the presence of population in the excited state can thereby be suppressed by this type of competition process. In particular, the efficiency of multiphoton ionization [4,22] harmonic generation [21] and parametric mixing processes [23] can be modified. In addition, since laser action requires a population inversion between two atomic levels, this interference effect can prevent the possibility of achieving laser action in schemes for laser-pumped lasers where a strong multiwave mixing process can occur [24]. Another consequence of the competition that has been discussed by Agarwal [25] is that the generated fields may represent a squeezed state of the field.

2. Theoretical investigation of competition between amplified spontaneous emission and four-wave mixing

In this section we present a theoretical treatment of the suppression of the upper level population due to the FWM process. We first solve to fourth order in perturbation theory the density matrix equations of motion for the three-level atomic system shown in fig. 2 in the presence of the incident and generated fields. We use the results of this calculation to determine the population in the upper level and to determine the nonlinear polarization of the medium and hence to derive coupled amplitude equations for the three interacting fields. We then study the nature of the solution to these equations and find that if the wavevector mismatch is not too large the fields evolve to reach spatially-invariant values and that in the presence of this spatially-invariant field the upper level population is at least partially suppressed. For the case of perfect phase matching, complete suppression of the upper level population is predicted.

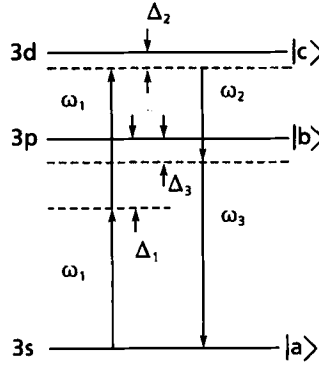


Fig. 2. Energy-level diagram showing the FWM process and the notation used in the calculation.

The nonlinear polarization of the medium can be obtained in terms of the density matrix elements as

$$P(z, t) = N(\mu_{ab}\rho_{ba}^{(3)} + \mu_{bc}\rho_{cb}^{(3)} + \text{c.c.}) \quad (1)$$

where N denotes the atomic number density. It is convenient to represent the nonlinear polarization in terms of its frequency components as

$$P(z, t) = \sum_j P(\omega_j) e^{i(k_j z - \omega_j t)} + \text{c.c.} \quad (j = 1, 2, 3). \quad (2)$$

The quantities $P(\omega_j)$ can be obtained from the solution of the density matrix equations of motion. We assume that the field incident upon the atomic vapor can be expressed as

$$E(z, t) = \sum_j \tilde{E}_j e^{-i\omega_j t} + \text{c.c.} \quad j = 1, 2, 3 \quad (3)$$

where $\tilde{E}_j = E_j \exp(-ik_j z)$. Our experiments were conducted under conditions such that to good approximation the inequalities

$$\Delta_1 \gg \Delta_3 \gg \Delta_2 \quad \text{and} \quad \Delta_1, \Delta_3 \gg \Gamma_{ba} \quad (4)$$

were satisfied where $\Delta_1 = \omega_{ba} - \omega_1$, $\Delta_2 = \omega_{ca} - 2\omega_1$, $\Delta_3 = \omega_{ba} - \omega_3$, and Γ_{ba} is the dipole dephasing rate of the $b \rightarrow a$ transition. Under these conditions, the nonlinear polarizations can be expressed as [3]

$$P(\omega_1) = 2\chi^{\text{FWM}} E_2 E_3 E_1^* e^{-i\Delta k z} + 2\chi_s^{\text{TPA}} |E_1|^2 E_1 \quad (5a)$$

$$P(\omega_2) = \chi^{\text{FWM}} E_1^2 E_3^* e^{+i\Delta k z} + \chi_d^{\text{TPA}} |E_3|^2 E_2 \quad (5b)$$

$$P(\omega_3) = \chi^{\text{FWM}} E_1^2 E_2^* e^{+i\Delta k z} + \chi_d^{\text{TPA}} |E_2|^2 E_3 \quad (5c)$$

where the susceptibilities for FWM and for two-photon absorption (TPA) involving the same and different frequencies are given by

$$\chi^{\text{FWM}} = \frac{\Delta_1}{\Delta_3} \chi_s^{\text{TPA}} = \frac{\Delta_3}{\Delta_1} \chi_d^{\text{TPA}} = \frac{N |\mu_{ba}|^2 |\mu_{cb}|^2}{\hbar^3 \Delta_1 \Delta_3 (\Delta_2 - i\Gamma_{ca})}, \quad (6)$$

and where $\Delta k = k_2 + k_3 - 2k_1$ is the wavevector mismatch and μ_{ij} denotes the ij matrix element of the dipole operator. The spatial evolution of the field amplitudes are described in the slowly varying amplitude approximation by the coupled amplitude equations

$$\frac{dE_j}{dz} = \frac{2\pi i \omega_j^2}{k_j c^2} P(\omega_j). \quad (7)$$

Through use of these equations, one can determine the variation of the intensity $S_j = (cn_j/2\pi) |E_j|^2$ of each field component as

$$\frac{dS_j}{dz} = -2\omega_j \text{Im}[E_j^* P(\omega_j)]. \quad (8)$$

Under this same set of conditions (4), the population in the upper level can be expressed as

$$\rho_{cc}^{(4)} = \frac{1}{\hbar N \gamma_c} \left[\text{Im}(\chi_s^{\text{TPA}}) |E_1|^4 + \text{Im}(\chi_d^{\text{TPA}}) |E_2|^2 |E_3|^2 + \text{Im}(\chi^{\text{FWM}}) (E_2 E_3 E_1^{*2} e^{-i\Delta k z} + \text{c.c.}) \right]. \quad (9)$$

Note that $\rho_{cc}^{(4)}$ contains an interference term which for appropriate values of the complex field amplitudes can cause the population in level c to vanish. This cancellation is the origin of the suppression of ASE by FWM. The origin of this cancellation can be traced to the expression for the second-order coherence between the ground and upper levels, which under the conditions can be written as

$$\rho_{ca}^{(2)} = \frac{\hbar \Delta_3}{N \mu_{bc} \mu_{ab}} \chi^{\text{FWM}} \left[E_1^2 + \frac{\Delta_1}{\Delta_3} E_2 E_3 e^{-i\Delta k z} \right] e^{i(2k_1 z - 2\omega_1 t)}. \quad (10)$$

When the complex field amplitudes obey the condition

$$E_1^2 = -\frac{\Delta_1}{\Delta_3} E_2 E_3 e^{-i\Delta k z} \quad (11)$$

$\rho_{ca}^{(2)}$ vanishes, and consequently all higher order contributions to the density matrix vanish. In particular, $\rho_{cc}^{(4)}$ vanishes, implying that no population is transferred to the upper level. Furthermore, the nonlinear polarization which is

proportional to $\rho_{cb}^{(3)}$ and to $\rho_{ba}^{(3)}$ also vanishes, and hence from eq. (7) one sees that the field amplitudes remain spatially invariant.

In order to solve the coupled amplitude equations (7), we rewrite them in terms of the real field amplitudes and phases. We express the complex field amplitudes as $E_j = A_j \exp(i\phi_j)$ and introduce the phase angle β of the nonlinear susceptibilities as

$$\tan \beta = \frac{\text{Im } i\chi^{\text{FWM}}}{\text{Re } i\chi^{\text{FWM}}} = -\frac{\Delta_2}{\Gamma_{ca}}. \quad (12)$$

In terms of these quantities, the coupled amplitude equations become

$$\frac{dA_1}{dz} = -\alpha_1 A_1 \left[A_2 A_3 \cos(\theta + \beta) + \frac{\Delta_3}{\Delta_1} A_1^2 \cos \beta \right] \quad (13a)$$

$$\frac{dA_2}{dz} = -\alpha_2 A_3 \left[A_1^2 \cos(\theta - \beta) + \frac{\Delta_1}{\Delta_3} A_2 A_3 \cos \beta \right] \quad (13b)$$

$$\frac{dA_3}{dz} = -\alpha_3 A_2 \left[A_1^2 \cos(\theta - \beta) + \frac{\Delta_1}{\Delta_3} A_2 A_3 \cos \beta \right] \quad (13c)$$

where the new phase variable $\theta = \phi_2 + \phi_3 - 2\phi_1 - \Delta kz$ obeys

$$\begin{aligned} \frac{d\theta}{dz} = & 2\alpha_1 A_2 A_3 \sin(\theta + \beta) + \left(\alpha_2 \frac{A_1^2 A_3}{A_2} + \alpha_3 \frac{A_1^2 A_2}{A_3} \right) \sin(\theta - \beta) \\ & + \left(2\alpha_1 \frac{\Delta_3}{\Delta_1} A_1^2 - \alpha_2 \frac{\Delta_1}{\Delta_3} A_3^2 - \alpha_3 \frac{\Delta_1}{\Delta_3} A_2^2 \right) \sin \beta - \Delta k, \end{aligned} \quad (14)$$

with coupling coefficients α_j given by

$$\alpha_1 = \frac{4\pi\omega_1^2}{k_1 c^2} |\chi^{\text{FWM}}|, \quad (15a)$$

$$\alpha_j = \frac{2\pi\omega_j^2}{k_j c^2} |\chi^{\text{FWM}}| \quad (j = 2, 3). \quad (15b)$$

In terms of the new variables A_j and θ , the population in the upper level is given by

$$\rho_{cc}^{(4)} = \frac{\text{Im } \chi^{\text{FWM}}}{N\hbar\gamma_c} \frac{\Delta_3}{\Delta_1} \left[A_1^4 + \left(\frac{\Delta_1}{\Delta_3} \right)^2 A_2^2 A_3^2 + 2 \frac{\Delta_1}{\Delta_3} A_2 A_3 A_1^2 \cos \theta \right], \quad (16)$$

and the condition for suppression under our experimental conditions where Δ_1 and Δ_3 have the same sign is $\theta = \pi$ and $A_2 A_3 / A_1^2 = \Delta_3 / \Delta_1$.

The set of coupled equations given by eqs. (13) and (14) are very difficult to solve in general. However, considerable simplification occurs for the special case in which the incident laser is tuned precisely to the two-photon-allowed transition so that $\Delta_2 = 0$ and hence $\beta = 0$ and in which the phase matching condition is precisely satisfied so that $\Delta k = 0$. In this case, eq. (14) has the form $d\theta/dz = f(z) \sin \theta$ where $f(z)$ is positive definite. One can see by inspection that the asymptotic solution of this equation for large values of z is $\theta \rightarrow \pi$. Physically, one expects that θ will approach this value after propagation through a distance several times the characteristic coupling distance $\Lambda = (\alpha_2 \alpha_3)^{-1/2} A_1^{-2}$. In this limit the coupled equations for the real amplitudes possess a steady-state solution in which the field amplitudes are spatially invariant and are related by

$$\frac{A_2 A_3}{A_1^2} = \frac{\Delta_3}{\Delta_1}. \quad (17)$$

This condition is precisely the condition which leads to the suppression of the upper level population. In order to determine the conditions under which the fields actually evolve so as to fulfill condition (17), one needs to solve the coupled set of equations (13). This set of equations does not yield a simple analytic solution. However, in our experiments the fractional conversion from the pump wave into the generated waves is always small which allows us to assume that the pump amplitude A_1 remains constant. In the constant pump limit the reduced set of equations can readily be solved analytically. The solution to the coupled equations (13b and c) can be written as $A_3 = C \cosh x$ and $A_2 = [(\alpha_2/\alpha_3)(A_3^2 - C^2)]^{1/2}$ where

$$x = \tanh^{-1} \left[\frac{1}{a} \left\{ (a^2 + b^2)^{1/2} \tanh \left((a^2 + b^2)^{1/2} z \right. \right. \right. \\ \left. \left. \left. + \tanh^{-1} \left[\frac{a \tanh x_0 + b}{(a^2 + b^2)^{1/2}} \right] \right) - b \right\} \right], \quad (18)$$

where x_0 and C are constants of integration, $a = (\alpha_2 \alpha_3)^{1/2} A_1^2$, and $b = (\alpha_2 \Delta_1 C^2 / 2 \Delta_3)$. In the limit $z \rightarrow \infty$ the ω_3 field approaches the value

$$A_3 \rightarrow A_1 \left(\frac{\Delta_3}{\Delta_1} \right)^{1/2} \left(\frac{\alpha_3}{\alpha_2} \right)^{1/4} \quad (19a)$$

and in the limit $C \rightarrow 0$

$$A_2 \rightarrow A_1 \left(\frac{\Delta_3}{\Delta_1} \right)^{1/2} \left(\frac{\alpha_2}{\alpha_3} \right)^{1/4} \quad \text{as } z \rightarrow \infty. \quad (19b)$$

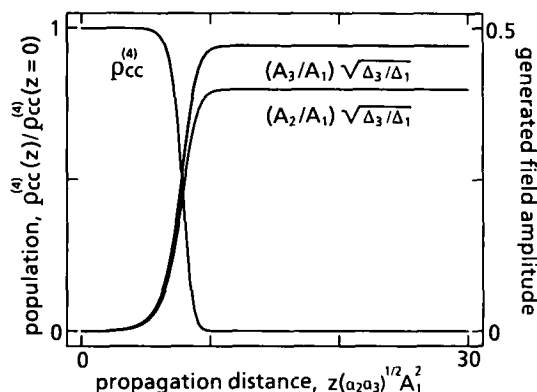


Fig. 3. Spatial evolution of the generated fields, A_2 and A_3 , and the excited state population, $\rho_{cc}^{(4)}$, for the case of perfect phase matching ($\Delta k = 0$) and exact tuning to the two-photon resonance ($\Delta_2 = 0$), as predicted by the theory described in the text.

The limit $C \rightarrow 0$ is the limit in which the ω_2 and ω_3 fields contain the same number of photons, and hence corresponds to the experimental situation in which these two fields grow from noise. Note that in the limit $C \rightarrow 0$, $z \rightarrow \infty$ the product of the asymptotic values of A_2 and A_3 approaches the value $A_1^2 \Delta_3 / \Delta_1$ as required by eq. (17) for a spatially-invariant solution and that, according to eq. (16) with $\theta = \pi$, $\rho_{cc}^{(4)}$ vanishes under these conditions. In fig. 3 we show the spatial evolution of A_2 and A_3 as given by eq. (18) for the case in which the constants of integration have the values $x_0 = 0$ and $C = 10^{-5} A_1$. Also shown at each position in the medium is the value of $\rho_{cc}^{(4)}$ calculated from eq. (16). Note that A_2 and A_3 grow to their steady-state values in several coupling lengths and that the upper level population is suppressed once the fields have attained these values.

In the more general case in which Δk and β are arbitrary, it is very difficult to solve the set of coupled equations (13 and 14) for the spatial evolution of the fields. However, in this case we can determine those conditions under which spatially-invariant solutions to these equations exist. We do this by setting all of the spatial derivatives to zero and solving the resulting set of equations algebraically. In general this set of equations does not possess a solution except for special values of Δk and β . However, in the constant pump limit (A_1 and ϕ_1 constant), which is relevant to our experimental conditions, spatially-invariant solutions for the ω_2 and ω_3 fields exist under fairly general conditions. In the constant pump limit the equations that must be solved simultaneously are eqs. (13b and c) and eq. (14) with the assumption that the A_1 field is constant which causes the terms that contain α_1 to drop out. We find that θ is given by the equation

$$\sin(\theta - \beta) + \tan \beta \cos(\theta - \beta) = \frac{\Delta k}{2(\alpha_2 \alpha_3)^{1/2} A_1^2} \quad (20a)$$

where for simplicity we take $C = 0$ since it is small compared to the steady-state values of A_2 and A_3 for the case of interest in which the fields grow from noise. The fields are then given in terms of θ by

$$A_2^2 = \frac{\alpha_2}{\alpha_3} A_3^2 = - \left(\frac{\alpha_2}{\alpha_3} \right)^{1/2} \frac{\Delta_3 A_1^2 \cos(\theta - \beta)}{\Delta_1 \cos \beta}. \quad (20b)$$

The fields and upper level population given by eq. (20) are shown for representative cases in fig. 4. In fig. 4a we show the relative phase angle θ of the optical fields plotted as a function of the normalized wavevector mismatch for several different values of the phase angle β characterizing the nonlinearity. Recall that only for $\theta = \pi$ can there be complete suppression of the upper-state population. Each curve is plotted over the entire domain of the values of Δk for which a spatially-invariant solution exists. For $\beta \neq 0$, this domain includes regions in which a weak input wave will not experience net gain and hence where the spatially-invariant solution cannot be reached by the growth of the generated fields from noise. Also in this region there are two steady-state solutions and both are shown on the figure. These regions cannot be realized in an experiment such as ours involving a single input frequency, and thus are shown dashed in the figure. In the limit of large detuning from the two-photon resonance, β approaches the value $\pi/2$ and θ attains the value π everywhere. Figure 4b shows the dependence of the field amplitude A_3 on the wavevector mismatch and on β . According to eq. (20), the field amplitude A_2 is directly proportional to A_3 . Figure 4c shows the dependence of the normalized upper-state population on Δk for various values of β . Note that the curves for different values of β fall on top of one another, except that they have different end points within the region where the curves are shown dashed and the two different solutions for the amplitude and phases shown in (a) and (b) lead to the same prediction for the excitation of population in level c . Curves of the sort shown in figs. 4a, b, and c can be obtained for negative values of β by reflecting the curves for positive values of β about the axis $\Delta k = 0$. For the conditions of our experiment, Δ_1 and Δ_3 are both positive, and hence, for collinear propagation of the three waves Δk is negative.

3. Experimental investigation of competition between amplified spontaneous emission and four-wave mixing

We have studied experimentally suppression of ASE by the four-wave mixing process by focusing the output of a tunable dye laser into a sodium vapor cell and detecting the emission from the cell in both the forward and backward directions. The dye laser operates at approximately 6855 Å and produces ~ 500 μJ of energy in a 0.7 cm^{-1} spectral bandwidth in a pulse duration of 1.5 to 10 ns.

The dye laser output is focused to a spot size of $\sim 100 \mu\text{m}$ in a sodium cell containing between 10^{15} and 10^{17} sodium atoms per cm^3 and ~ 10 torr of helium buffer gas with a length of approximately 5 cm. The ASE and FWM processes have different spatial and spectral signatures, which allow the processes to be

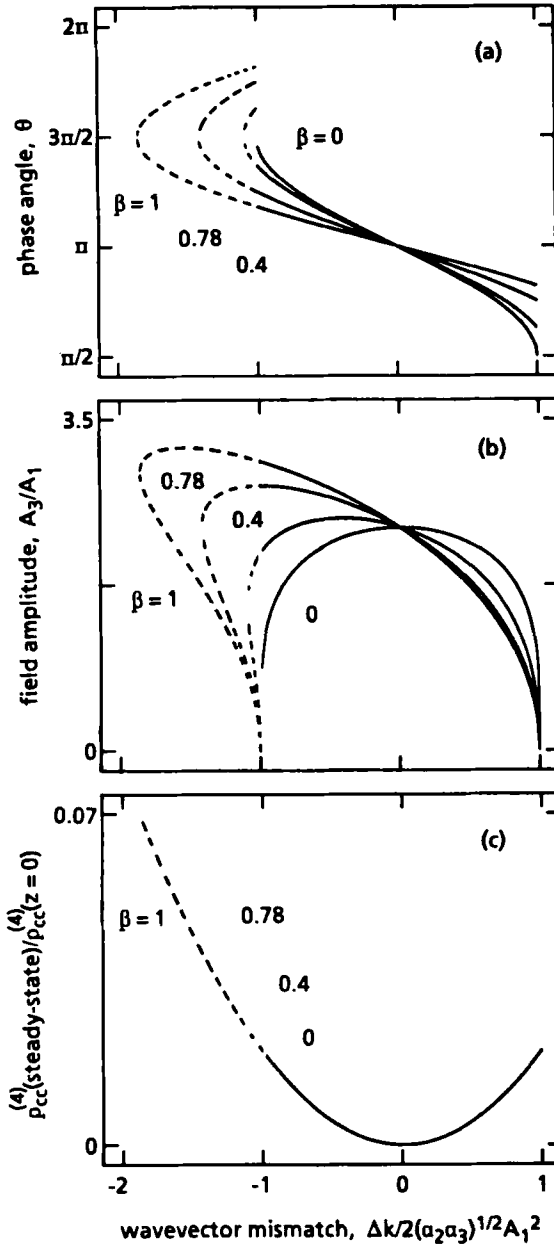


Fig. 4. (a) and (b) Spatially-invariant solutions to the coupled amplitude equations in the constant-pump limit, plotted as a function of the normalized wavevector mismatch for several different values of the phase angle β of the nonlinear susceptibility. The solution is plotted over the entire range of values of Δk for which a stable, spatially-invariant solution exists. (c) Upper level population $\rho_{cc}^{(4)}$ in the presence of the fields shown in (a) and (b), normalized by its value in the presence of the pump field alone. All of these curves lie on top of one another; however, curves corresponding to different values of β have different end points as indicated. The dashed portions of the curves correspond to conditions under which the solution will not grow from noise.

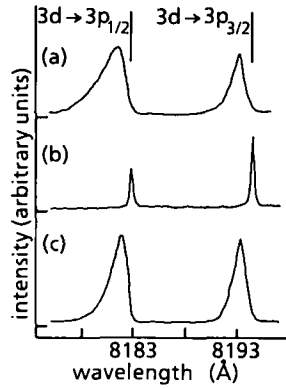


Fig. 5. (a) Spectrum of the radiation generated by FWM in the forward direction for a detuned laser. (b) Spectrum of the radiation (ASE) emitted in the backward direction. (c) Spectrum of the radiation emitted in the forward direction with the laser tuned precisely to the two-photon resonance. Note that this spectrum is that of FWM and not that of ASE.

unambiguously distinguished. Due to phasematching requirements FWM can occur only in the forward direction, and for the number densities and detunings used in our experiment this radiation has a shape of a cone surrounding the transmitted laser beam [28]. When ASE occurs it has equal gain the forward and backward directions, and has the shape of a filled-in disk. FWM and ASE can also be distinguished spectrally. In order to determine the spectral composition of the light, the radiation is imaged onto the entrance slit of a 0.75-m spectrometer having a resolution of 0.2 Å. The portion of the output spectrum that is emitted is a cone in the forward direction near the $3d \rightarrow 3p$ transition frequency is shown in fig. 5a. The spectrum is seen to be composed of two components due to resonance enhancement by the two fine structure components of the 3p level. Each component extends from resonance to higher frequencies because the phase matching condition for FWM can be met only for noncollinear propagation for this sign of the detuning [28]. ASE can occur in the backward direction when the laser is tuned to the exact two-photon resonance and when the suppression of the upper level population is not complete and fig. 5b shows the emission spectra in the backward direction under these conditions. The emission is weak compared to the emission in the forward direction and is seen to consist of two narrow components centered at the two transition frequencies. Figure 5c shows the spectrum of the forward emission when the laser is tuned to the exact two-photon resonance. The spectrum is seen to be that of FWM and not of ASE, even though the calculated gain for FWM is much less than that for ASE calculated in the absence of competition. In particular, the gain coefficient for the FWM process is given for $N = 5 \times 10^{16}$ and $A_1 = 80$ esu by $\Lambda^{-1} = 33 \text{ cm}^{-1}$. The gain cross section

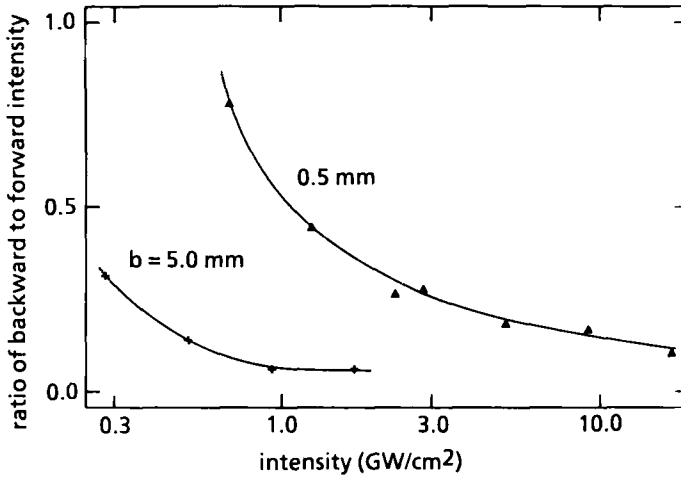


Fig. 6. Ratio of the intensity in the backward direction to that in the forward direction versus incident laser intensity for two different values of the beam confocal parameter. The limiting values of unity and zero correspond to pure ASE and pure FWM respectively.

at line center for the $3d \rightarrow 3p$ transition under conditions of Doppler broadening is $5 \times 10^{-12} \text{ cm}^2$, and the calculated population inversion ignoring competition effects is $5 \times 10^{15} \text{ cm}^3$, leading to a calculated gain coefficient for ASE of $2.5 \times 10^4 \text{ cm}^{-1}$.

In order to quantify the degree to which FWM suppresses ASE, we have measured the ratio of the emission near the $3d \rightarrow 3p$ transition frequency in the backward direction (which is solely ASE) to that in the forward direction (which is due to both ASE and FWM). Shown in fig. 6 is a plot of this ratio as a function of the peak value of the laser intensity for two different values of the pump beam confocal parameter b . The sodium number density is $\sim 1 \times 10^{16} \text{ cm}^{-3}$. It is seen that FWM suppresses ASE more efficiently when the laser intensity and hence the gain for FWM is highest. Since FWM is a phase-matched process, the confocal parameter of the pump laser is an appropriate measure of the effective interaction length. The suppression is seen to be more complete when the interaction occurs over a longer portion of the cell that is associated with the use of a larger confocal parameter.

Figure 7 shows the conversion efficiency of the pump wave into the ω_2 wave in both the forward and backward directions as a function of the position of the focus of the incident laser beam relative to the center of the 5 cm cell. These data were collected at a sodium number density of $1 \times 10^{16} \text{ cm}^{-3}$, a confocal parameter of 5 mm, and a peak laser intensity of 400 MW cm^{-3} . Note that the emission into the forward direction is maximized when the focal spot is near the center of the cell. However, the emission into the backward direction (ASE) is suppressed

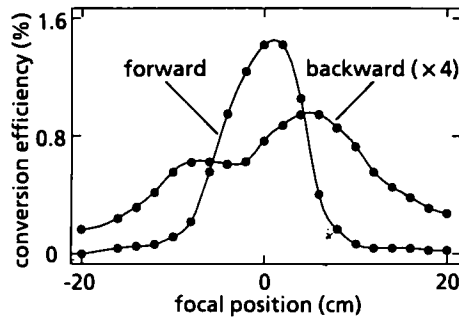


Fig. 7. Intensities of the emission in the forward and backward directions versus position of the beam waist in the cell. Note that ASE is suppressed when FWM is enhanced.

when the position of the focus is near the position that maximizes the forward emission. The dip in the curve is actually shifted slightly toward the entrance face of the cell because under this condition the initiation of the FWM process can occur closer to the entrance face and hence the FWM process can suppress the upper level population over a greater fraction of the length of the cell.

In summary, we have investigated the competition between four-wave mixing and amplified spontaneous emission both theoretically and experimentally and have shown that four-wave mixing due to the two-photon-resonant response of an atomic vapor can lead to the suppression of amplified spontaneous emission from the excited level. The suppression is due to an interference between two excitation pathway connecting the ground and excited states. Under conditions of perfect phase matching, the fields generated by the four-wave mixing process evolve in such a way that there is complete destructive interference between these two pathways, and hence no population is placed in the upper level. As the wavevector mismatch is increased, the degree of suppression of the upper level population becomes less complete. We have observed this suppression experimentally and have shown that the suppression does not occur when the medium is excited by counterpropagating laser beams that cannot efficiently excite the FWM process.

We acknowledge fruitful discussions with K. Rzazewski. This work was supported by the U.S. Army Research Office.

References

- [1] W. Kaiser and M. Maier, *Laser Handbook*, eds. F.T. Arecchi and E.O. Schulz-Dubois (North-Holland, Amsterdam, 1972).
- [2] M.S. Malcuit, D.J. Gauthier and R.W. Boyd, *Phys. Rev. Lett.*, 55 (1985) 1086.
- [3] R.W. Boyd, M.S. Malcuit, D.J. Gauthier and K. Rzazewski, *Phys. Rev. A* 35 (1987) 1648.

- [4] J.C. Miller, R.N. Compton, M.G. Payne and W.W. Garrett, *Phys. Rev. Lett.* 45 (1980) 114.
- [5] J.H. Glowina and R.K. Sander, *Phys. Rev. Lett.* 49 (1982) 12.
- [6] K. Aron and P.M. Johnson, *J. Chem. Phys.* 76 (1977) 5099.
- [7] F.H.M. Faisal, R. Wallenstein and H. Zacharias, *Phys. Rev. Lett.* 39 (1977) 1138.
- [8] D.J. Jackson and J.J. Wynne, *Phys. Rev. Lett.* 49 (1982) 543.
- [9] D.J. Jackson, J.J. Wynne and P.H. Kes, *Phys. Rev. A* 28 (1983) 781.
- [10] J.J. Wynne, *Phys. Rev. Lett.* 52 (1984) 751.
- [11] G.S. Agarwal and S.P. Tewari, *Phys. Rev. A* 29 (1984) 1922.
- [12] M.G. Payne and W.R. Garrett, *Phys. Rev. A* 26 (1982) 356.
- [13] D. Normand, J. Morellec and J. Reif, *J. Phys. B* 16 (1983) L227.
- [14] R.N. Compton and J.C. Miller, *J. Opt. Soc. Am. B* 2 (1985) 355.
- [15] S.P. Tewari and G.S. Agarwal, *Phys. Rev. Lett.* 56 (1986) 1811.
- [16] M.G. Payne, W.R. Garrett, and W.R. Ferrell, *Phys. Rev. A* 34 (1986) 1143.
- [17] W.R. Garrett, W.R. Ferrell, M.G. Payne and J.C. Miller, *Phys. Rev. A* 34 (1986) 1165.
- [18] S.M. Hamadani, J.A.D. Stockdale, R.N. Compton and M.s. Pindzola, *Phys. Rev. A* 34 (1986) 1938.
- [19] V.V. Krasnikov, M.S. Pshenichnikov and V.S. Solomatin, *JETP Lett.* 43 (1986) 149.
- [20] E.A. Manykin and A.M. Afanas'ev, *Sov. Phys. JETP* 25 (1976) 828.
- [21] H. Kildal and S.R.J. Brueck, *IEEE J. Quantum Electron.* QE-16 (1980) 556.
- [22] J. Zhanh, Q. Li, L. Zhao and Y. Ni, *Chinese Phys. Lett.* 2 (1985) 453.
- [23] J. Bokor, R.R. Freeman, R.L. Panock and J.C. White, *Opt. Lett.* 6 (1981) 182.
- [24] R.R. Freeman, J. Bokor and W.E. Cooke, *Phys. Rev. A* 236 (1982) 3029.
- [25] G.S. Agarwal, *Phys. Rev. Lett.* 57 (1986) 827.
- [26] R. Karplus and J. Schwinger, *Phys. Rev.* 73 (1948) 1020.
- [27] N. Bloembergen and Y.R. Shen, *Phys. Rev.* 133 (1964) A37.
- [28] J. Krasinski, D.J. Gauthier, M.S. Malcuit and R.W. Boyd, *Opt. Commun.* 54 (1985) 241.